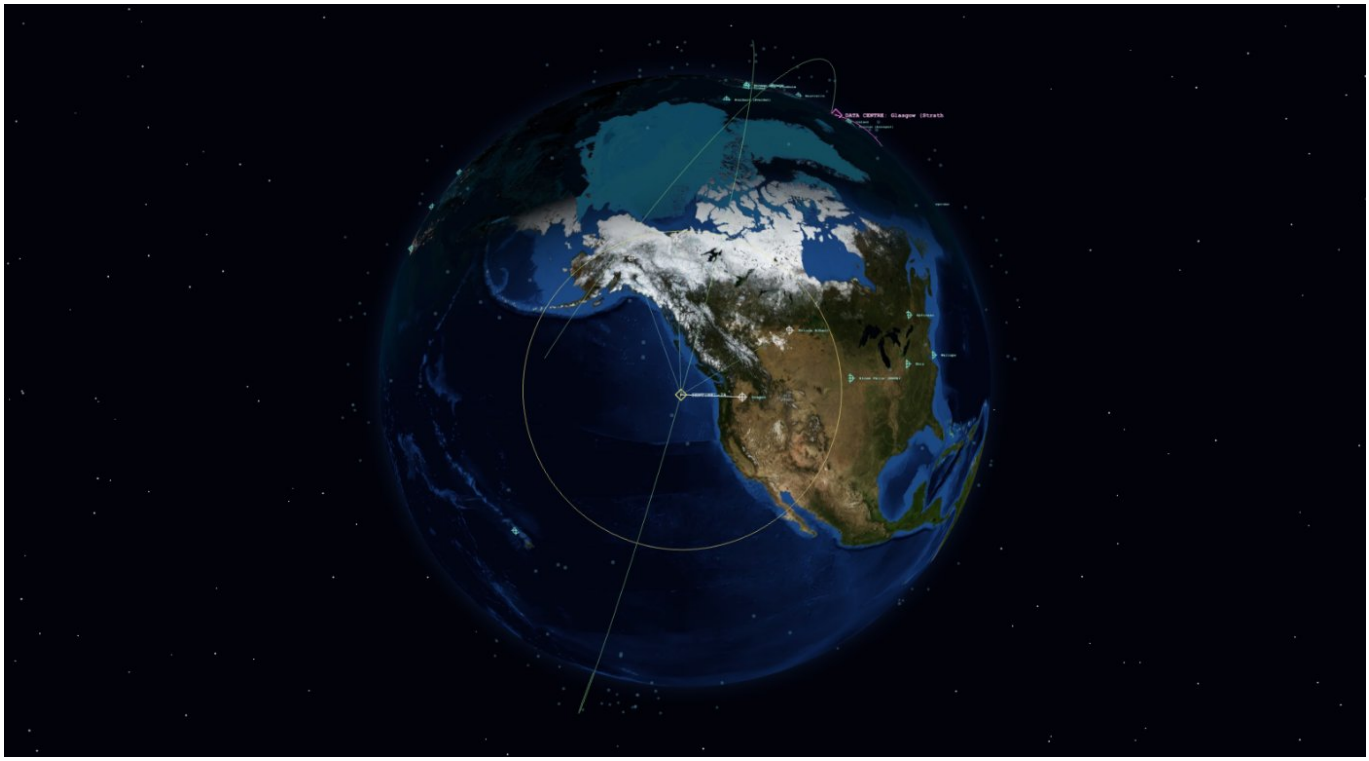
Earth-observation link report

SENTINEL-2A -> Glasgow (Strathclyde)

Generated 2026-11-06 19:51:49 UTC. Simulation clock 2026-11-06 19:51:49 UTC. Payload: Frog fibres, one frame (EO sim) (159 bit). Station network: all (39 sites with X-band, elevation mask 5 deg). No target: the message is created on board now.



Globe at the simulation time: selected satellite (yellow diamond) with its footprint at the elevation mask and ground track; stations (cyan, white when in view); data centre (magenta) with the fibre arc from the receiving station; target (orange); GEO relays (yellow, far out).

1. The satellite

item	value
Name / NORAD / COSPAR	SENTINEL-2A / 40697 / 2015-028A
Family / operator	Sentinel-2 / ESA / Copernicus
Sensor / resolution / swath	MSI, 13 bands 443–2190 nm; 10 / 20 / 60 m; 290 km
Position at simulation time	lat 47.101 deg, lon -131.137 deg, alt 795.6 km, 7.461 km/s
Orbit	incl 98.562 deg, period 100.58 min (14.317 rev/day), perigee 785 km, apogee 787 km, e 0.000115
Elements	epoch 2026-11-06 14:12:45 UTC (age 5.7 h); CelesTrak GP data, SGP4 propagation
TLE line 1	1 40697U 15028A 26310.59219456 .00000407 00000+0 17197-3 0 9993
TLE line 2	2 40697 98.5623 23.3544 0001154 95.0520 265.0795 14.30820062594091

Payload downlink (SOURCED)

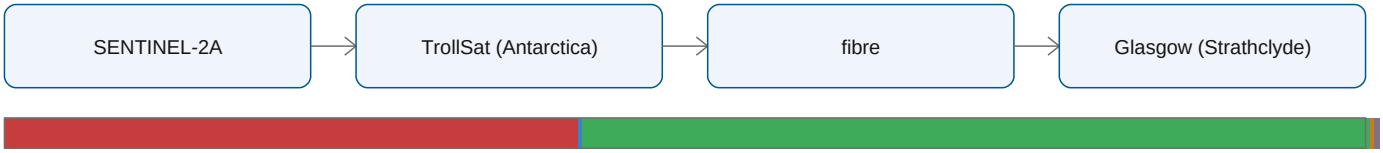
parameter	value
Band / centre frequency	X-band, 8.0900 GHz
Nominal data rate	520.0 Mbit/s
Modulation / coding	8PSK; 3 bit/symbol, code rate 0.667; required Eb/N0 6.5 dB
Satellite EIRP	24 dBW (ASSUMED - not published)
TT&C channel	S-band 2250 MHz, 2.0 Mbit/s
Direct broadcast	none
GEO relay	EDRS, 1.80 Gbit/s - Laser terminal to EDRS; near-real-time delivery to Harwell / Weilheim / Redu gateways
Sources	ESA Sentinel-2 mission: 520 Mbit/s X-band downlink, EDRS 1.8 Gbit/s | ESA EDRS

2. The communication path and where the time goes

Message creation time t0 = 2026-11-06 19:51:49 UTC: message created on board now. Payload 159 bit (eo_satellite_sim SUMMARY.md: 159 bit/frame). Every route below is computed for this payload from the 48 h contact prediction; a payload larger than one pass can carry is drained over successive passes.

On-board queue: 93.8 GB already in the recorder ahead of the payload, drained first-in-first-out at the payload rate (mission volume 1500 GB/day (eoPortal Sentinel-2: 1.5 TB/day after compression, 2.4 Tbit memory)). Transmitter duty cycle: 30 min per orbit on the payload link. The TT&C route uses the 43 S-band stations of the network, has no recorder queue and spends no payload-transmitter time.

Direct downlink to the station network: 41 min 34 s



term	time	share	note
wait for a pass over the network	17 min 32 s	42.2 %	6 passes needed
radio propagation (slant range)	6.87 ms	0.0 %	2061 km at closest approach
recorder backlog drained ahead of the payload (FIFO)	24 min 2 s	57.8 %	93.8 GB queued
serialisation at 520 Mbit/s	0.3 µs	0.0 %	one CCSDS frame minimum (1,115 B)
fibre TrollSat (Antarctica) -> Glasgow (Strathclyde)	112.25 ms	0.0 %	14228 km great circle × 1.5
ground processing / forwarding	0.0 µs	0.0 %	

Total 41 min 34 s; the payload arrives at 2026-11-06 20:33:23 UTC after 6 passes. Space-segment rate 520.0 Mbit/s.

GEO relay (EDRS): 33 min 39 s



term	time	share	note
wait for line of sight to EDRS-C	33 min 39 s	100.0 %	
LEO -> GEO propagation	126.04 ms	0.0 %	~37786 km
GEO -> gateway propagation	130.71 ms	0.0 %	39187 km to Harwell (EDRS gateway)
serialisation at 1800 Mbit/s	5.0 µs	0.0 %	
fibre Harwell (EDRS gateway) -> Glasgow (Strathclyde)	9.84 ms	0.0 %	514 km
ground processing / forwarding	0.0 µs	0.0 %	

Total 33 min 39 s; the payload arrives at 2026-11-06 20:25:28 UTC. Space-segment rate 1.80 Gbit/s.

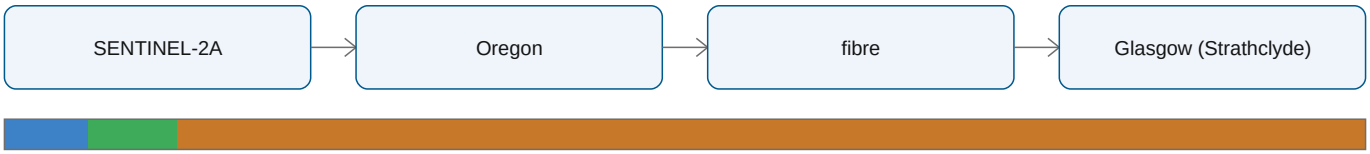
Own antenna at the data centre (hypothetical): 4 h 39 min



term	time	share	note
wait for a pass over Glasgow (Strathclyde)	4 h 15 min	91.4 %	3 passes
radio propagation	3.52 ms	0.0 %	1054 km
recorder backlog drained ahead of the payload (FIFO)	24 min 2 s	8.6 %	93.8 GB queued
serialisation at 520 Mbit/s	0.3 µs	0.0 %	
ground processing / forwarding	0.0 µs	0.0 %	

Total 4 h 39 min; the payload arrives at 2026-11-07 00:30:47 UTC after 3 passes. Space-segment rate 520.0 Mbit/s.

S-band TT&C channel (semantic message in telemetry): 68.29 ms



term	time	share	note
wait for a pass over any S-band station (43)	0.0 µs	0.0 %	next: Oregon
radio propagation	4.18 ms	6.1 %	1254 km
serialisation at 2.0 Mbit/s (telemetry channel)	4.46 ms	6.5 %	no recorder queue, no X-band transmitter time
fibre Oregon -> Glasgow (Strathclyde)	59.64 ms	87.3 %	7183 km
ground processing / forwarding	0.0 µs	0.0 %	

Total 68.29 ms; the payload arrives at 2026-11-06 19:51:49 UTC. Space-segment rate 2.0 Mbit/s.

The payload on the radio

quantity	value
Information bits	159 bit
Channel bits actually sent	8.9 kbit (CCSDS AOS frames of 1115 bytes; one frame minimum)
Serialisation at 520.0 Mbit/s	0.3 µs for the payload bits; 17.2 µs framed
P(at least one bit error), coded link	1.59e-10
P(at least one bit error), uncoded QPSK	0.00e+0
Same payload on the S-band TT&C channel (2.0 Mbit/s)	4.46 ms

Reading: for a semantic message of a few hundred bits the radio time is microseconds and the fibre tens of milliseconds; the delay is decided by whether a station or a relay is in view when the message is created. The frog link gains nothing from a faster radio and everything from an always-on low-rate channel.

3. Link budget and noise

Computed for the link now, to Oregon (AWS Ground Station).

quantity	value	how
Frequency	8.0900 GHz	catalogue
Slant range / elevation	1256.0 km / 35.06 deg	SGP4 position vs station
One-way propagation delay	4.190 ms	range / c
Doppler shift	-29.10 kHz	range-rate 1.079 km/s x f / c
Satellite EIRP	24.0 dBW (assumed)	catalogue
Free-space path loss	-172.59 dB	32.45 + 20 log f[MHz] + 20 log d[km]
Atmospheric gases	-0.09 dB	ITU-R P.676 zenith value / sin(el)
Rain 0 mm/h	-0.00 dB	ITU-R P.838 k R^alpha x effective path (3 km rain height)
Pointing + polarisation	-0.8 dB	0.5 + 0.3 dB assumed
Ground antenna 5.4 m	+50.99 dBi	10 log(eta (pi D / lambda)^2), eta 0.6
G/T (Tsys 150 K)	29.23 dB/K	G - 10 log Tsys
Received carrier power	-98.48 dBW (141.870 pW)	EIRP + G - losses
Noise density N0 = k Tsys	-206.84 dBW/Hz	k = 1.380649e-23 J/K
Occupied bandwidth	351.00 MHz	260.00 Msym/s x (1 + 0.35 roll-off)
Noise power in band	-121.39 dBW	N0 + 10 log B
C/N0	108.36 dB-Hz	EIRP + G/T - losses + 228.6
SNR in band	22.90 dB	C - N
Eb/N0 at 520.0 Mbit/s	21.20 dB	C/N0 - 10 log Rb

quantity	value	how
Required Eb/N0 + implementation	8.00 dB	8PSK, r=0.67 (CCSDS 130.1-G) + 1.5 dB
LINK MARGIN	13.20 dB (OK)	
BER uncoded QPSK	1.46e-59	0.5 erfc(sqrt(Eb/N0))
BER coded (model)	1.00e-12	1e-6 at the requirement, one decade per 0.3 dB
Shannon capacity in this band	2.67 Gbit/s	B log2(1 + SNR)
Nominal rate / capacity	19.5 %	

4. Contacts in the next 48 h

Stations in view now (EL >= 5 deg): Oregon EL 35.1 deg, 1256 km; Prince Albert EL 12.6 deg, 2172 km; Fairbanks (ASF) EL 8.5 deg, 2473 km; Inuvik EL 6.7 deg, 2620 km. GEO relay: not in line of sight.

AOS (UTC)	LOS	station	network	dur	max EL	min range	AZ AOS->LOS
11-06 19:51:47	19:52:13	Inuvik	CSA / DLR / SSC	0.4 min	7 deg	2609 km	175 -> 177
11-06 19:51:47	19:52:43	Fairbanks (ASF)	NASA NEN	0.9 min	9 deg	2464 km	145 -> 150
11-06 19:51:47	19:53:44	Prince Albert	CSA / NRCan	1.9 min	13 deg	2164 km	260 -> 247
11-06 19:51:47	19:57:11	Oregon	AWS Ground Station	5.4 min	35 deg	1254 km	283 -> 225
11-06 19:55:56	20:03:58	Hawaii (Oahu)	AWS Ground Station / NOAA	8.0 min	12 deg	2211 km	55 -> 137
11-06 20:21:30	20:34:17	McMurdo	NASA / NOAA	12.8 min	87 deg	818 km	47 -> 224
11-06 20:30:25	20:39:50	TrollSat (Antarctica)	KSAT	9.4 min	15 deg	2061 km	160 -> 63
11-06 20:40:53	20:52:04	Cape Town	AWS Ground Station	11.2 min	26 deg	1523 km	147 -> 21
11-06 20:42:03	20:54:35	Hartebeesthoek	SANSA	12.5 min	75 deg	824 km	165 -> 351
11-06 21:00:39	21:13:07	Matera	e-GEOS / ASI (Copernicus core)	12.5 min	69 deg	845 km	173 -> 343
11-06 21:02:38	21:14:24	Toulouse (Aussaguel)	CNES	11.8 min	38 deg	1194 km	138 -> 358
11-06 21:04:14	21:16:43	Neustrelitz	DLR	12.5 min	66 deg	861 km	173 -> 342
11-06 21:05:44	21:17:25	Ireland	AWS Ground Station	11.7 min	36 deg	1242 km	131 -> 353
11-06 21:05:55	21:18:02	Stockholm	AWS Ground Station	12.1 min	42 deg	1117 km	187 -> 338
11-06 21:08:28	21:20:15	Kiruna (ESTRACK)	ESA	11.8 min	34 deg	1299 km	197 -> 337
11-06 21:08:29	21:20:15	Esrange (Kiruna)	SSC	11.8 min	33 deg	1303 km	197 -> 337
11-06 21:08:56	21:20:51	Tromsø	KSAT	11.9 min	36 deg	1242 km	194 -> 337
11-06 21:11:21	21:23:20	Svalbard (SvalSat)	KSAT	12.0 min	37 deg	1220 km	192 -> 337
11-06 21:19:39	21:32:06	Inuvik	CSA / DLR / SSC	12.5 min	67 deg	863 km	25 -> 221
11-06 21:21:08	21:33:42	Fairbanks (ASF)	NASA NEN	12.6 min	82 deg	806 km	21 -> 199
11-06 21:25:33	21:26:45	Prince Albert	CSA / NRCan	1.2 min	5 deg	2773 km	333 -> 321
11-06 21:28:59	21:29:45	Oregon	AWS Ground Station	0.8 min	5 deg	2776 km	317 -> 310
11-06 21:33:24	21:45:30	Hawaii (Oahu)	AWS Ground Station / NOAA	12.1 min	48 deg	1023 km	1 -> 209
11-06 22:01:29	22:14:06	McMurdo	NASA / NOAA	12.6 min	55 deg	971 km	12 -> 210
11-06 22:10:11	22:22:12	TrollSat (Antarctica)	KSAT	12.0 min	35 deg	1286 km	158 -> 16
11-06 22:20:08	22:31:36	Cape Town	AWS Ground Station	11.5 min	31 deg	1365 km	184 -> 316
11-06 22:23:59	22:30:12	Hartebeesthoek	SANSA	6.2 min	9 deg	2462 km	216 -> 276
11-06 22:38:35	22:50:31	Maspalomas	ESA / INTA	11.9 min	42 deg	1106 km	147 -> 0
11-06 22:42:23	22:53:43	Toulouse (Aussaguel)	CNES	11.3 min	28 deg	1441 km	200 -> 331
11-06 22:44:54	22:57:04	Ireland	AWS Ground Station	12.2 min	44 deg	1082 km	186 -> 338
11-06 22:45:07	22:49:16	Matera	e-GEOS / ASI (Copernicus core)	4.2 min	6 deg	2640 km	254 -> 294
11-06 22:46:54	22:55:00	Neustrelitz	DLR	8.1 min	12 deg	2234 km	236 -> 318
11-06 22:49:28	22:56:18	Stockholm	AWS Ground Station	6.8 min	9 deg	2414 km	251 -> 318
11-06 22:51:30	23:00:21	Tromsø	KSAT	8.9 min	14 deg	2120 km	244 -> 335
11-06 22:51:30	22:59:29	Kiruna (ESTRACK)	ESA	8.0 min	11 deg	2263 km	251 -> 331
11-06 22:51:31	22:59:29	Esrange (Kiruna)	SSC	8.0 min	11 deg	2266 km	251 -> 331
11-06 22:52:54	23:03:32	Svalbard (SvalSat)	KSAT	10.6 min	21 deg	1714 km	231 -> 348

AOS (UTC)	LOS	station	network	dur	max EL	min range	AZ AOS->LOS
11-06 22:59:06	23:10:31	Inuvik	CSA / DLR / SSC	11.4 min	31 deg	1363 km	31 -> 260
11-06 23:00:46	23:12:26	Fairbanks (ASF)	NASA NEN	11.7 min	35 deg	1261 km	21 -> 244
11-06 23:30:22	23:42:11	Dubbo	AWS Ground Station	11.8 min	37 deg	1224 km	38 -> 179
11-06 23:42:22	23:54:09	McMurdo	NASA / NOAA	11.8 min	31 deg	1404 km	334 -> 199
11-06 23:49:57	00:02:43	TrollSat (Antarctica)	KSAT	12.8 min	81 deg	823 km	153 -> 334
11-07 00:19:24	00:28:46	Maspalomas	ESA / INTA	9.4 min	16 deg	1969 km	215 -> 314
11-07 00:29:27	00:34:11	Ireland	AWS Ground Station	4.7 min	7 deg	2610 km	260 -> 305
11-07 00:32:45	00:33:47	Gatineau	NRCan	1.0 min	5 deg	2772 km	52 -> 42
11-07 00:35:26	00:44:01	Svalbard (SvalSat)	KSAT	8.6 min	13 deg	2174 km	272 -> 359
11-07 00:37:59	00:48:19	Inuvik	CSA / DLR / SSC	10.3 min	21 deg	1713 km	46 -> 294
11-07 00:40:06	00:49:46	Fairbanks (ASF)	NASA NEN	9.7 min	18 deg	1884 km	27 -> 285
11-07 00:51:35	01:03:22	Hatoyama (EOC)	JAXA	11.8 min	36 deg	1240 km	27 -> 169
11-07 00:53:23	01:01:08	Seoul	AWS Ground Station	7.8 min	11 deg	2268 km	51 -> 129
11-07 01:07:46	01:20:18	Alice Springs	Geoscience Australia	12.5 min	78 deg	813 km	17 -> 190
11-07 01:10:31	01:21:01	Dubbo	AWS Ground Station	10.5 min	21 deg	1727 km	332 -> 218
11-07 01:12:11	01:21:27	Dongara	SSC	9.3 min	16 deg	1988 km	65 -> 161
11-07 01:24:20	01:34:26	McMurdo	NASA / NOAA	10.1 min	18 deg	1920 km	294 -> 188
11-07 01:29:31	01:41:54	TrollSat (Antarctica)	KSAT	12.4 min	51 deg	1018 km	144 -> 296
11-07 01:39:52	01:45:52	Punta Arenas	SSC / KSAT / AWS	6.0 min	8 deg	2532 km	130 -> 72
11-07 01:45:06	01:53:42	Córdoba	CONAE	8.6 min	13 deg	2147 km	131 -> 43
11-07 01:47:41	01:59:36	Cuiabá	INPE	11.9 min	39 deg	1168 km	152 -> 7
11-07 02:03:49	02:14:50	Wallops	NASA NEN	11.0 min	27 deg	1487 km	130 -> 5
11-07 02:05:26	02:17:09	Gatineau	NRCan	11.7 min	36 deg	1227 km	136 -> 357
11-07 02:05:54	02:15:21	Ohio	AWS Ground Station	9.5 min	17 deg	1918 km	112 -> 12
11-07 02:10:58	02:15:00	Sioux Falls (EROS)	USGS Landsat	4.0 min	6 deg	2647 km	70 -> 32
11-07 02:12:51	02:19:23	Prince Albert	CSA / NRCan	6.5 min	9 deg	2415 km	73 -> 9
11-07 02:16:05	02:26:32	Inuvik	CSA / DLR / SSC	10.4 min	22 deg	1682 km	71 -> 317
11-07 02:18:29	02:27:01	Fairbanks (ASF)	NASA NEN	8.5 min	14 deg	2118 km	44 -> 317
11-07 02:18:31	02:25:00	Svalbard (SvalSat)	KSAT	6.5 min	9 deg	2464 km	311 -> 14
11-07 02:30:50	02:42:54	Miyun	CRESDA / RADI	12.1 min	42 deg	1113 km	24 -> 173
11-07 02:30:59	02:43:26	Seoul	AWS Ground Station	12.5 min	75 deg	820 km	10 -> 200
11-07 02:31:21	02:42:05	Hatoyama (EOC)	JAXA	10.7 min	24 deg	1575 km	352 -> 233
11-07 02:37:37	02:48:58	Sanya	CRESDA / RADI	11.4 min	30 deg	1389 km	33 -> 166
11-07 02:42:54	02:53:38	Singapore	KSAT / AWS	10.7 min	24 deg	1596 km	42 -> 162
11-07 02:49:43	03:01:57	Dongara	SSC	12.2 min	47 deg	1042 km	357 -> 203
11-07 02:51:42	02:55:48	Alice Springs	Geoscience Australia	4.1 min	6 deg	2648 km	293 -> 254
11-07 03:07:14	03:15:02	McMurdo	NASA / NOAA	7.8 min	11 deg	2345 km	253 -> 177
11-07 03:08:40	03:20:24	TrollSat (Antarctica)	KSAT	11.7 min	34 deg	1320 km	129 -> 263
11-07 03:17:24	03:29:49	Punta Arenas	SSC / KSAT / AWS	12.4 min	49 deg	1025 km	159 -> 2
11-07 03:22:56	03:35:17	Córdoba	CONAE	12.3 min	56 deg	942 km	175 -> 333
11-07 03:28:19	03:37:07	Cuiabá	INPE	8.8 min	14 deg	2073 km	207 -> 297
11-07 03:42:46	03:54:29	Wallops	NASA NEN	11.7 min	34 deg	1279 km	193 -> 333
11-07 03:43:21	03:55:46	Ohio	AWS Ground Station	12.4 min	63 deg	880 km	175 -> 342

Passes over the data centre itself (Glasgow (Strathclyde)): 11-06 21:06 (46 deg, 12 min); 11-06 22:45 (37 deg, 12 min); 11-07 00:30 (6 deg, 4 min); 11-07 09:11 (9 deg, 6 min); 11-07 10:49 (46 deg, 12 min); 11-07 12:29 (38 deg, 12 min); 11-07 14:09 (11 deg, 7 min); 11-07 19:01 (9 deg, 6 min); 11-07 20:36 (31 deg, 11 min); 11-07 22:15 (60 deg, 12 min); 11-07 23:58 (12 deg, 8 min); 11-08 10:20 (29 deg, 11 min); 11-08 11:59 (59 deg, 12 min); 11-08 13:39 (16 deg, 9 min); 11-08 18:33 (6 deg, 3 min).

GEO relay line-of-sight windows: EDRS-C 11-06 20:25 - 21:25; EDRS-A 11-06 20:26 - 21:26; EDRS-C 11-06 22:04 - 23:05; EDRS-A 11-06 22:06 - 23:05; EDRS-C 11-06 23:41 - 00:49; EDRS-A 11-06 23:45 - 00:46; EDRS-C 11-07 01:10 - 03:37; EDRS-A 11-07 01:21 - 02:31; EDRS-A 11-07 02:46 - 05:15; EDRS-C 11-07 03:56 - 05:04; EDRS-A 11-07 05:37 - 06:44; EDRS-C 11-07 05:40 - 06:41.

5. Same routes, every payload

payload	bits	direct X-band	GEO relay	own antenna	S-band TT&C
Frog Class-4 alarm message (event flag + region)	64 bit	41 min 34 s (6 p)	33 min 39 s	4 h 39 min (3 p)	68.29 ms
Frog fibres, one frame (EO sim)	159 bit	41 min 34 s (6 p)	33 min 39 s	4 h 39 min (3 p)	68.29 ms
Frog fibres, one frame (CDnet @25 fps)	321 bit	41 min 34 s (6 p)	33 min 39 s	4 h 39 min (3 p)	68.29 ms
Frog fibres, 5-min observation (300 frames)	48.0 kbit	41 min 34 s (6 p)	33 min 39 s	4 h 39 min (3 p)	87.83 ms
Frog fibres + evidence crops, 5 min	109.0 kbit	41 min 34 s (6 p)	33 min 39 s	4 h 39 min (3 p)	118.33 ms
JPEG q50 frame 320×240	23.4 kbit	41 min 34 s (6 p)	33 min 39 s	4 h 39 min (3 p)	75.54 ms
H.264 frame, 8 kbit/frame setting	8.2 kbit	41 min 34 s (6 p)	33 min 39 s	4 h 39 min (3 p)	68.29 ms
JPEG q50, 5-min observation	7.03 Mbit	41 min 34 s (6 p)	33 min 39 s	4 h 39 min (3 p)	3.58 s
Raw 8-bit frame 320×240	614.4 kbit	41 min 34 s (6 p)	33 min 39 s	4 h 39 min (3 p)	371.03 ms
Sentinel-2 L1C tile (100 km², ~600 MB)	4.80 Gbit	41 min 43 s (6 p)	33 min 42 s	4 h 39 min (3 p)	58 min 34 s (8 p)
Landsat 8/9 L1 scene (~1 GB)	8.00 Gbit	41 min 50 s (6 p)	33 min 44 s	4 h 39 min (3 p)	1 h 33 min (20 p)
WorldView-3 pan+MS strip, 13 × 100 km (~4 GB)	32.00 Gbit	42 min 36 s (7 p)	33 min 57 s	4 h 40 min (3 p)	5 h 58 min (50 p)
Sentinel-1 IW SLC product (~8 GB)	64.00 Gbit	43 min 37 s (7 p)	34 min 15 s	4 h 41 min (3 p)	11 h 44 min (100 p)

6. Methods, assumptions and sources

Orbits: CelesTrak GP element sets (US Space Force catalogue) propagated with SGP4 (satellite.js). Accuracy about 1 km at the element epoch, degrading with age. Positions and pass times were cross-checked against the independent Python sgp4 package (agreement < 0.06 deg, pass times within 10 s) and against Heavens-Above (same passes; it uses a 10 deg mask).

Contacts: 48 h scan at 20 s with AOS/LOS refined by bisection to under 1 s, above the chosen elevation mask. Target opportunities use the elevation equivalent to the maximum off-nadir angle: cos(el) = sin(eta) (R + h) / R. GEO relay line of sight: the LEO-GEO segment must clear the Earth by 60 km.

Delivery time = waiting for the next contact (multi-pass drain for large files, at the nominal rate) + propagation (range / c) + serialisation (payload bits in CCSDS AOS frames of 1,115 bytes, one frame minimum, at the nominal rate) + terrestrial fibre (great-circle distance x 1.5 route factor at 2/3 c plus 0.5 ms per hop for 12 hops) + the ground processing time entered. GEO relay: LEO-GEO about 37,800 km, GEO-gateway from geometry, relay rate from the catalogue, fibre from the gateway.

Link budget: free-space loss, ITU-R P.676 gaseous absorption (zenith value / sin el), ITU-R P.838 rain (k, alpha at the band; 3 km rain height; P.618 reduction), pointing 0.5 dB, polarisation 0.3 dB, implementation 1.5 dB, antenna efficiency 0.6, roll-off 0.35, required Eb/N0 from CCSDS 130.1-G performance curves. Uncoded BER = 0.5 erfc(sqrt(Eb/N0)); coded BER is a waterfall model (1e-6 at the requirement, one decade per 0.3 dB, floor 1e-12), indicative only.

Why real missions quote hours (docs/LATENCY_RESEARCH.md): a satellite has about 10 minutes of ground contact per 100-minute orbit and 4-6 windows a day with any one station; it collects 0.5-1.5 TB/day into its recorder, so imagery queues for hours behind earlier data and the X-band transmitter is duty-cycle limited (Sentinel-1: 30 min per orbit); then the product is processed under a timeliness policy (Sentinel NRT 1-3 h, standard 24 h; Landsat ~5 h). The simulator models the queue (backlog, FIFO or priority), the duty cap and the processing time; a semantic message avoids the queue and the payload transmitter entirely on the TT&C route.

What is NOT modelled: station scheduling and contention, encryption and product processing, data policy (e.g. ESA's near-real-time commitment for Sentinel products is under 3 h, set by processing and policy rather than the radio), antenna slew and acquisition time (typically 1 to 2 minutes at the start of a pass), and the actual availability of relay service.

Confidence tags in the catalogue: SOURCED = operator or engineering source (listed on the card and in section 1); TYPICAL = secondary source (eoPortal, trade press), treat as +/-20 %; ASSUMED = class default. Satellite EIRP and the ground G/T are never published per mission and are always assumptions, so the margin and BER are indicative rather than a certified budget.

Payload sizes: the frog link, JPEG and H.264 figures are measured in this project's simulations (eo_satellite_sim, E8 held-out run); full product sizes from the operators (Copernicus, USGS, Maxar).

Stations used in this report

station	network	lat	lon	bands	dish	source
Svalbard (SvalSat)	KSAT	78.23	15.39	S/X/Ka	13 m	KSAT: 78°N, 160+ antennas, up to 14 passes/day for polar orbits
Tromsø	KSAT	69.66	18.94	S/X/Ka	11 m	KSAT Tromsø
TrollSat (Antarctica)	KSAT	-72.01	2.54	S/X/Ka	7.3 m	KSAT TrollSat 72°S
Inuvik	CSA / DLR / SSC	68.32	-133.55	S/X	13 m	Inuvik Satellite Station Facility
Kiruna (ESTRACK)	ESA	67.86	20.97	S/X	15 m	ESA: KIR-2 67.858428°N 20.966880°E, S/X-band
Esrange (Kiruna)	SSC	67.88	21.07	S/X/Ka	13 m	SSC Esrange
Fairbanks (ASF)	NASA NEN	64.86	-147.85	S/X	11 m	Alaska Satellite Facility
McMurdo	NASA / NOAA	-77.85	166.67	S/X/Ka	10 m	NASA McMurdo Ground Station
Neustrelitz	DLR	53.33	13.07	S/X	7.3 m	DLR Neustrelitz
Matera	e-GEOS / ASI (Copernicus core)	40.65	16.70	X	10 m	Matera 40°382573N 16°422153E, Sentinel X-ban station
Maspalomas	ESA / INTA	27.76	-15.63	S/X	15 m	ESA: 15 m antenna 27°452463N 15°382023W
Sioux Falls (EROS)	USGS Landsat	43.73	-96.62	S/X	10 m	USGS EROS Landsat Ground Station
Wallops	NASA NEN	37.94	-75.47	S/X	11 m	NASA Wallops
Hartebeesthoek	SANSA	-25.89	27.69	S/X	12 m	SANSA Hartebeesthoek
Punta Arenas	SSC / KSAT / AWS	-53.00	-70.85	S/X	7.3 m	SSC Punta Arenas; AWS Ground Station
Alice Springs	Geoscience Australia	-23.76	133.88	S/X	9 m	GA Alice Springs (Landsat / Sentinel)
Dongara	SSC	-29.05	115.35	S/X	7.3 m	SSC Western Australia Space Centre
Dubbo	AWS Ground Station	-32.24	148.60	S/X	5.4 m	AWS Ground Station Dubbo
Singapore	KSAT / AWS	1.35	103.82	S/X	7.3 m	KSAT Singapore; AWS Ground Station
Dubai	KSAT	25.20	55.30	S/X	7.3 m	KSAT Dubai
Bahrain	AWS Ground Station	26.00	50.60	S/X	5.4 m	AWS Ground Station Bahrain
Mauritius	KSAT	-20.30	57.60	S/X	7.3 m	KSAT Mauritius
Hawaii (Oahu)	AWS Ground Station / NOAA	21.30	-158.00	S/X	5.4 m	AWS Ground Station Hawaii
Oregon	AWS Ground Station	45.90	-119.30	S/X	5.4 m	AWS Ground Station Oregon
Ohio	AWS Ground Station	40.40	-82.90	S/X	5.4 m	AWS Ground Station Ohio
Ireland	AWS Ground Station	53.30	-6.30	S/X	5.4 m	AWS Ground Station Ireland
Stockholm	AWS Ground Station	59.30	18.10	S/X	5.4 m	AWS Ground Station Stockholm
Seoul	AWS Ground Station	37.50	127.00	S/X	5.4 m	AWS Ground Station Seoul
Cape Town	AWS Ground Station	-33.90	18.40	S/X	5.4 m	AWS Ground Station Cape Town
Prince Albert	CSA / NRCan	53.21	-105.93	S/X	10 m	Prince Albert Satellite Station
Gatineau	NRCan	45.58	-75.81	S/X	10 m	Gatineau Satellite Station
Shadnagar (NRSC)	ISRO	17.03	78.18	S/X	10 m	ISRO NRSC Shadnagar
Hatoyama (EOC)	JAXA	36.00	139.35	S/X	10 m	JAXA Earth Observation Center
Miyun	CRESDA / RADI	40.40	116.90	S/X	12 m	RADI Miyun station
Kashi	CRESDA / RADI	39.50	76.00	S/X	12 m	RADI Kashi station
Sanya	CRESDA / RADI	18.30	109.30	S/X	12 m	RADI Sanya station
Cuiabá	INPE	-15.55	-56.07	S/X	11 m	INPE Cuiabá
Córdoba	CONAE	-31.52	-64.46	S/X	7.3 m	CONAE Córdoba
Toulouse (Aussaguel)	CNES	43.43	1.50	S/X	10 m	CNES Aussaguel

Verification links: https://www.n2yo.com/satellite/?s=40697 ; https://www.heavens-above.com/PassSummary.aspx?satid=40697&lat=55.8600&lng=-4.2400&alt=0&tz=UCT&showAll=t ; https://celestrak.org/NORAD/elements/gp.php?CATNR=40697&FORMAT=tle